Sensitivity of a calibrated SLD observable

$N = 10, \quad B/J = 0.40, \quad h = 0, \quad h_{\mathrm{ref}} = 0.001$

QFI, F_h — Matched SLD, $\mathcal{M}[\hat{L}_h]$ — Calibrated SLD, $\mathcal{M}[\hat{L}_{h_{\mathrm{ref}}}]$

(a) $\left(\left|E_1\right\rangle + \left|E_{\bar{1}}\right\rangle\right)/\sqrt{2}$

(b) $\left|E_1\right\rangle$

(c) $\left|\uparrow\cdots\uparrow\right\rangle$

(d) $\left(\left|\uparrow\cdots\uparrow\right\rangle + \left|\downarrow\cdots\downarrow\right\rangle\right)/\sqrt{2}$

$\frac{2\pi}{\Delta_{2\bar{2}}}$ $\frac{2\pi}{\Delta_{1\bar{1}}}$

$\mathcal{M}/(Nt)^2$, $F_h/(Nt)^2$,

0.00 0.08 0.16 0.24 0.32

10^0 10^1 10^2 10^3 10^4

t/T

$\mathcal{M}[\hat{O}] = (\partial_h\langle\hat{O}\rangle)^2/\mathrm{Var}(\hat{O})$, with $\hat{O}$ held fixed under ∂_h